CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B.Tech Examination (IT)

March-Apr 2018

IT302.02/IT302.01/IT302 Design & Analysis of Algorithm

Date: 31.03.2018, Saturday **Time: 01:30 p.m. to 04:30 p.m.** **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

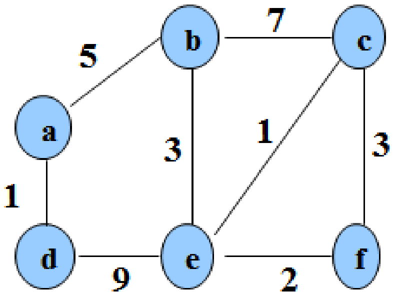
SECTION-I

Q-1 Attempt Following. **[07]**

1. Explain general recurrence relation of divide and conquer technique. [03]
2. Give the difference between Greedy Approach and Dynamic Programming Approach. [02]
3. Find out total number of primitive operations for following code. [02]

```
A_max(A,n)
{
    max=A[0]
    for(i=1;i<n;i++)
{
    if(A[i]>max)
        max=A[i];
}
return max
}
```

Q-2 (a) Find the minimum spanning tree using kruskal's algorithm. **[06]**



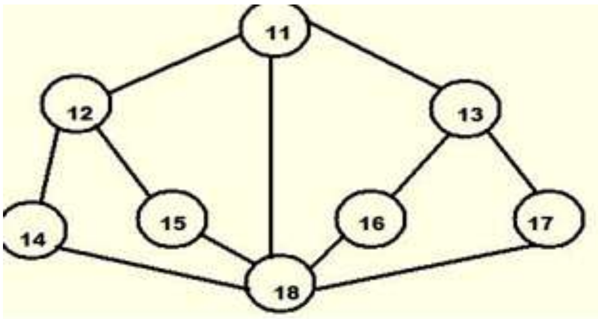
Q-2 (b) What is an algorithm? How to analyze time and space complexity of algorithm? Explain with suitable example. **[04]**

Q-2 (c) Let S= {1, 2, …, n} is set of jobs and each job i has a deadline $d_i \geq 0$ and a profit $p_i \geq 0$. One unit of time to process each job and we can do at most one job each time. Earn the profit p_i if job i is completed by its deadline. Solve above job scheduling problem and solve the following problem using algorithm. **[04]**

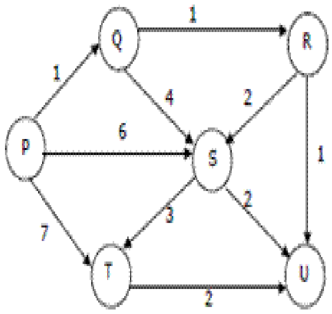
I	1	2	3	4	5
p_i	20	15	10	5	1
d_i	2	2	1	3	3

OR

Q-2 (b) Define Graph. Apply DFS and BFS graph traversal techniques on given graph with source vertex 11. **[04]**



Q-2 (c) Apply Dijkstra's single source shortest path algorithm on the following edge weighted directed graph with vertex P as the source vertex. Also note down the sequence of the vertex visited. **[04]**



Q-3 (a) Explain characteristics of greedy algorithm. **[04]**

(b) **Attempt the following questions. (Any Two)** **[10]**

1. Give similarities and differences between backtracking and branch-and-bound. Draw pruned state space tree for 4 queen problem.
2. Solve the following fractional Knapsack Problem using Greedy Algorithm. There are five items whose weights and values are given in following arrays.
   Weight w[] = { 10,14,12,7,15 } Value v[] = { 20,35,25,40,30 } Show your equation and find out the optimal knapsack items for weight capacity of 35 units.
3. Use branch and bound to solve the assignment problem (Minimization) with the following cost matrices.

	J1	**J2**	**J3**	**J4**
P1	11	12	18	40
P2	14	15	13	22
P3	11	17	19	23
P4	17	14	20	28

SECTION-II

Q-4 **Attempt following.** **[07]**

1. Define Time Complexity. Explain with the help of a suitable example Best case, Worst case and Average case time complexity of an algorithm. [03]

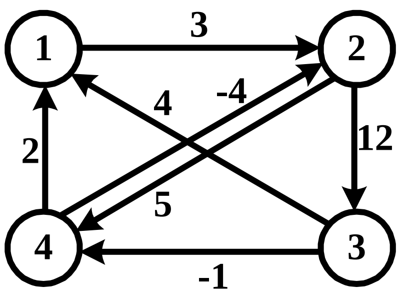
2. What is Asymptotic Notation? Explain Big Oh (O) notation with a suitable example. [02]

3. Construct the table for calculating Binomial Coefficient for Compute (n, k) = (7,5) [02]

Q-5 (a) Solve the knapsack problem to get optimal profit using dynamic programming. number of objects are 4, w(i)=(1,2,3,3),v(i)=(10,12,15,14) and knapsack capacity is 7 weight. **[04]**

Q-5 (b) Discuss basic idea of Rabin-Karp string matching algorithm. What is the running time complexity of this algorithm? **[05]**

Q-5 (c) A directed weighted graph is given as below. Find all pair shortest path using Floyd Algorithm. **[05]**



OR

Q-5 (b) State Master Theorem and Solve following recurrence using Master Theorem. **[05]**

1. $T(n) = 4T(n/2) + n$
2. $T(n) = 4T(n/2) + n^2$

Q-5 (c) Define the given terms with example: P Class, Np Class, Np Complete Class, NP Hard Class. Also Show the relationship between four classes. **[05]**

Q-6 Attempt the following questions. (Any Two) [14]

1. What is the use of Recurrence Tree? Solve the given recurrence using recurrence Tree method. T(n)=3T(n/3)+n.
2. Using Dynamic programming, find an optimum order of a matrix chain product whose sequence of dimensions is <5, 10, 3, 5, 6, 2>.
3. Define Longest common subsequence problem. Find the length of the subsequence for X (0 1 1 1 0 1 1) and Y (1 1 1 1 0 0 1). Also write algorithms for finding longest common subsequence.

****